



Open Ended Lab – Worksheet

Name:	ID:	Section:
-------	-----	----------

i. State Transition Diagram

Sketch the state transition diagram. Indicate the input conditions that cause transitions among states. Use the signals 'left' and 'right' for the direction indicator.

ii. State Assignment

Along with your state transition diagram, pick any state assignment method to map symbolic states to binary values.



iii. State Transition Table

Construct state transition table based on state diagram and chosen state assignment. Add columns for mapping the states to output.



--

State Table

	Current State			Reset	Right	Left	Next State		
	A	B	C	RES	R	L	AN	BN	CN
1	0	0	0				0	0	0
2	0	0	0						
3	0	0	1						
4	0	1	0						
5	0	1	1						
6	0	0	0						
7	1	0	0						
8	1	0	1						
9	1	1	0						
10	0	0	0						
11	x	x	X						
12	1	1	1						

	Current State			Lights (Outputs)					
	A	B	C	Ra	Rb	Rc	La	Lb	Lc
1	0	0	0	0	0	0	0	0	0
2	0	0	0						
3	0	0	1						
4	0	1	0						
5	0	1	1						
6	0	0	0						
7	1	0	0						
8	1	0	1						
9	1	1	0						
10	0	0	0						
11	1	1	1						

iv. *Circuit Diagram of FSM*

Obtain input and output equations and sketch circuit diagram to implement the FSM (a sample of Mealy FSM is assuming the D-flip flop as memory element is given in Figure 12.1 as a reference)

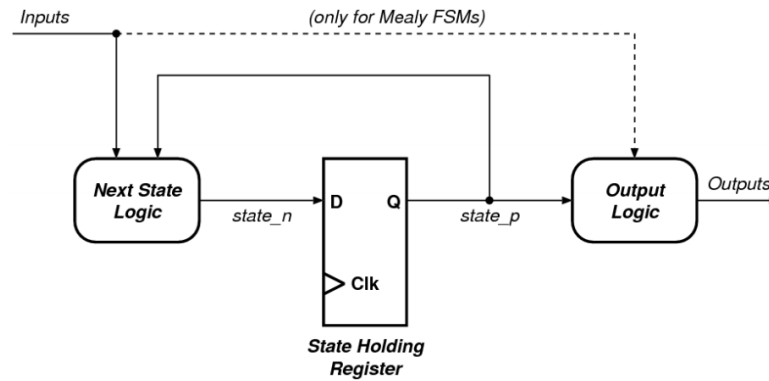


Figure 12.1 - A simple view of an FSM



v. ***Verilog Implementation***

On Eda Playground or Vivado create a new project Add a new Verilog source and implement the finite state machine using what you have learned in the lecture about Verilog. Be sure to include a clock input (clk) and a reset (rst) in your circuit.

Provide appropriately **commented** code here

vi. ***Implementing the Clock***

The clock available on Basys-3 board is 100 MHz, means that every clock period is 10 nanoseconds long. The entire blinking sequence would be then 40 ns. This is a very short time. light travels less than 250 meters during that time. If we want to see our sequence, we need to find a way to slow down the circuit. This can be achieved in two ways. We can either implement a clock divider circuit that divides the clock by a few million times, or we can generate an enable signal every few million cycles, and then use this enable signal to control our next state transition. Refer to Lab 10 to design a clock divider of your desired value



Provide appropriately **commented** code of clock div

vii. Top Level Module

Write top module to implement FSM along with clock divider



Provide appropriately **commented** code for designed module



viii. Define the constraints (to create .xdc file)

Define constraints for your module to implement on Basys-3 Board

Add Constraint file here



ix. Test on BASYS 03 Board

Program Basys-3 board to test and validate your designed module

Show your results to the RA, or else this will not be graded.



Assessment Rubric Open Ended Lab Finite State Machine

Name:	Student ID:
-------	-------------

Marks Distribution:

	Task No.	OE3 Code	OE5 Results	OE2 Design	OE5 Report
In - Lab	Task i	/10	-		/10
	Task ii	/10	-		
	Task iii	/10	-		
	Task iv	-	-	/10	
	Task v	/10	-		
	Task vi	/10	-		
	Task vii	/05	-		
	Task viii	/05	-		
	Task ix	-	/10		
Total: 100		/60	/10	/10	/10
CLO Mapping		CLO3	CLO3	CLO3	CLO4

Affective Domain Rubric		Points	CLO Mapped
AR 7	Report Submission and Viva	/10	CLO 4

CLO	Total Points	Points Obtained
3	80	
4	20	
Total	100	



Open-Ended Lab Evaluation Assessment Rubric

#	Assessment Elements	Level 1: Unsatisfactory	Level 2: Developing	Level 3: Good	Level 4: Exemplary
OE1	Problem Definition	Fails to clearly define the problem or research question.	The problem has been defined to some extent but lacks precision or relevance.	The problem or research question has been defined clearly and contextually	The problem has been defined clearly/ Research has been done with precision and relevance.
OE2	Experimental Design	The proposed design is unsubstantiated or does not satisfy most of the given constraints. The breakdown of system into features is incomplete and still at lower resolution.	The justification for the proposed design is weak, and it only satisfies some constraints. The breakdown of the system is missing some features and resolution is not appropriate in some places.	The proposed design is substantiated, preferably through proper analysis, and satisfies most given constraints. The breakdown of system into features seems complete, but resolution is not appropriate at some places.	The proposed design is substantiated, preferably through proper analysis, and satisfies all given constraints. The breakdown of system into features seems complete and at appropriate resolution, given students' level.
OE3	Data Collection	Measurements are incomplete, inaccurate and imprecise. Observations are incomplete or not included. Symbols, units and significant figures are not included.	Measurements are somewhat inaccurate and imprecise. Observations are incomplete or vague. Major errors are there in the use of symbols, units and significant digits.	Measurements are mostly accurate. Observations are generally complete. Minor errors are present in using symbols, units and significant digits.	Measurements are both accurate and precise. Data collection is systematic. Observations are very thorough and include appropriate symbols, units and significant digits and tasks completed in due time.
OE4	Analysis & Interpretation	Fails to analyze data effectively or draw meaningful interpretations.	The analysis of data is incomplete or important trends have been missed or provided limited interpretations.	The data has been analyzed effectively, by identifying evident trends and drawing reasonable interpretations.	In depth analysis has been made by identifying trends, patterns, and relationships. Insightful interpretations have been drawn.
OE5	Results, Plots and report	Figures/ graphs / tables are not developed or are poorly constructed with erroneous results. Titles, captions, units are not mentioned. Data is presented in an obscure manner or not presented at all.	Figures, graphs and tables are drawn but contain errors. Titles, captions, units are not accurate. Data presentation is not too clear. Most of the requirements are included in the report but are not well explained.	All figures, graphs, tables are correctly drawn but contain minor errors or some of the details are missing. Report meets all prescribed requirements.	Figures / graphs / tables are correctly drawn, and appropriate titles/captions and proper units are mentioned. Data presentation is systematic. Report meets all requirements, and it is prepared in original and creative way to engage readers.
OE6	Reflection	Lacks meaningful self-evaluation or reflection on	Offers limited self-evaluation without	Reflects on the experiment, identifying	Demonstrates critical self-evaluation, identifying



		the experiment.	significant insights.	areas of success and areas for improvement.	strengths, weaknesses, and areas for improvement in the experimental approach.
--	--	-----------------	-----------------------	---	--